PHYS 6031
Spring 2024
Homework 8
Due: Friday 29 March 2024 at 4:00 pm in my Clippinger mailbox

1. Answer the following items regarding plane-wave propagation at boundaries between materials:

(a) Fill in the steps necessary to go from Eq. (7.37) of Jackson to Eq. (7.39).

(b) Fill in the steps necessary to go from Eq. (7.37) of Jackson to Eq. (7.41).

(c) Show that when $\mu = \mu'$ (which is true for light propagation involving many optical materials together with air), these results simplify to the Fresnel Equations:

$\vec{\boldsymbol{E}} \parallel$ to plane of incidence	$\vec{\boldsymbol{E}} \perp$ to plane of incidence
$\frac{E_o''}{E_o} = \frac{\tan(i-r)}{\tan(i+r)}$	$\frac{E_o''}{E_o} = \frac{\sin(r-i)}{\sin(r+i)}$
$\frac{E_o'}{E_o} = \frac{2\cos i \sin r}{\sin(i+r)\cos(i-r)}$	$\frac{E_o'}{E_o} = \frac{2\cos i \sin r}{\sin(i+r)}$

2. Suppose space is divided into three regions, with three different indices of refraction:

$$\begin{array}{ll} z < 0 & n = n_1 \\ 0 \le z \le d & n = n_2 \\ z > d & n = n_3. \end{array}$$

assume that an incident plane wave is traveling in the $+z$ direction (i.e., coming from $z = -\infty$). The magnetic permeability is the same in all three media. Determine the transmission and reflection coefficients. In this case, the transmission coefficient is calculated for the transmitted intensity into region 3.

3. Consider a plane wave of frequency ω that is normally incident on a material of dielectric constant ϵ, magnetic permeability $\mu = \mu_0$, and conductivity σ. Recall that a coductivity of σ implies that $\vec{\boldsymbol{J}} = \sigma\vec{\boldsymbol{E}}$. In this case you will need to consider the wavevector $\vec{\boldsymbol{k}}$ to be a complex quantity.

(a) Determine the amplitude and phase of the reflected wave.

(b) Show that in the limit of a good conductor (very large σ), the reflection coefficient is given by $R = 1 - 2\omega\delta/c$, where δ is the skin depth defined by $\delta = \sqrt{2/\mu\sigma\omega}$.